

When descriptive forecast-horizon gains do not imply global evidence of incremental meteorological predictability: a multi-site PM10 study

Stochastic Environmental Research and Risk Assessment

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Online Resource 1

Complete site–horizon DM–HLN results for the fixed paired loss-differential ledger.

Definitions and multiplicity. The paired loss differential is $d_t = L_{\text{lags}} - L_{\text{lags+meteorology}}$, so positive statistics favour the lags + meteorology information set. The primary specification uses $q_{\text{overlap}} = 0$; the sensitivity uses the fixed Bartlett/Newey–West bandwidth $q = \lfloor 4(n/100)^{2/9} \rfloor$. The global Bonferroni family contains 36 planned comparisons, and Benjamini–Hochberg adjustment is applied across the four horizons within each station. No cell survives global Bonferroni correction. Non-significance is not evidence of equivalence.

Online Resource 1: Complete cell-level inferential results.

Site	<i>h</i>	<i>n</i>	Primary $q_{\text{overlap}} = 0$				Automatic Newey–West				<i>P</i>	<i>P</i> Bonf	<i>P</i> BH	BH
			DM–HLN	<i>p</i>	<i>P</i> Bonf	<i>P</i> BH	DM–HLN	<i>p</i>	<i>P</i> Bonf	<i>P</i> BH				
Madrid Casa de Campo	1	354	1.16901717167	0.243185513423	1	0.461127872171	No	5	1.09444286438	0.274507007573	1	0.499712256018	No	
Madrid Casa de Campo	6	356	-0.0501284771506	0.960048198833	1	0.960048198833	No	5	-0.0502606382815	0.959942956776	1	0.959942956776	No	
Madrid Casa de Campo	12	346	2.65199344286	0.0083712676589	0.30136563572	0.0334850706356	Yes	5	2.45461931349	0.0145966537132	0.525479533676	0.0583866148529	No	
Madrid Casa de Campo	24	354	0.943940024099	0.345845904128	1	0.461127872171	No	5	0.888672640586	0.374784192014	1	0.499712256018	No	
Birr (Co. Offaly)	1	211	2.68182334315	0.00790448604525	0.284561497629	0.031617944181	Yes	4	2.76482664258	0.00620211969451	0.223276309002	0.024808478778	Yes	
Birr (Co. Offaly)	6	212	-0.549510966919	0.583235962218	1	0.583235962218	No	4	-0.532490924927	0.594946325488	1	0.594946325488	No	
Birr (Co. Offaly)	12	212	0.889722415746	0.374628228749	1	0.583235962218	No	4	0.805852206852	0.421235321327	1	0.594946325488	No	
Birr (Co. Offaly)	24	212	0.709758952342	0.478637240291	1	0.583235962218	No	4	0.717731243685	0.473716890067	1	0.594946325488	No	
Dublin Airport	1	145	1.63629890787	0.103960822112	1	0.41584328845	No	4	2.06763867474	0.0404637959258	1	0.161855183703	No	
Dublin Airport	6	145	-0.272343290744	0.785748193811	1	0.785748193811	No	4	-0.280021116996	0.779863152556	1	0.779863152556	No	
Dublin Airport	12	145	-0.442456464142	0.658822789151	1	0.785748193811	No	4	-0.435781764352	0.663647630671	1	0.779863152556	No	
Dublin Airport	24	145	0.791336911172	0.4300489168	1	0.785748193811	No	4	1.0368665915	0.301536043364	1	0.603072086728	No	
Dundalk (Co. Louth)	1	151	1.11683202723	0.265851901536	1	0.37932160474	No	4	1.13751543125	0.25713663346	1	0.319920528826	No	
Dundalk (Co. Louth)	6	152	0.916937270381	0.360637671982	1	0.37932160474	No	4	0.997907784974	0.319920528826	1	0.319920528826	No	
Dundalk (Co. Louth)	12	153	0.881717781539	0.37932160474	1	0.37932160474	No	4	1.13209696598	0.259376459853	1	0.319920528826	No	
Dundalk (Co. Louth)	24	152	1.44794605704	0.149705275419	1	0.37932160474	No	4	1.34571761301	0.180410726124	1	0.319920528826	No	
Pearse St. Dublin	1	212	1.84652176009	0.066217009037	1	0.0882893453827	No	4	1.87250686726	0.0625197018369	1	0.0833596024492	No	
Pearse St. Dublin	6	212	2.65868458856	0.00844602734838	0.304056984542	0.0337841093935	Yes	4	2.94898656562	0.00354760341693	0.127713723009	0.0141904136677	Yes	
Pearse St. Dublin	12	212	1.87089930681	0.0627433356344	1	0.0882893453827	No	4	2.51309318992	0.0127154842108	0.457757431589	0.0254309684216	Yes	
Pearse St. Dublin	24	212	0.897446999464	0.370503297296	1	0.370503297296	No	4	0.923197738495	0.356958657325	1	0.356958657325	No	
Ringsend Dublin	1	212	1.23326014644	0.218850848114	1	0.842040952723	No	4	1.22664694816	0.221322349742	1	0.782338913659	No	
Ringsend Dublin	6	212	-0.227336153603	0.820382482283	1	0.924552049737	No	4	-0.197206527154	0.843855758549	1	0.93579706645	No	
Ringsend Dublin	12	212	0.806225655438	0.421020476362	1	0.842040952723	No	4	0.859265039691	0.391169456829	1	0.782338913659	No	
Ringsend Dublin	24	212	0.0948146258121	0.924552049737	1	0.924552049737	No	4	0.0806495962773	0.93579706645	1	0.93579706645	No	
Edenderry (Co. Offaly)	1	208	0.339187278463	0.734812504307	1	0.775205969578	No	4	0.335712343794	0.737427809091	1	0.806936687333	No	
Edenderry (Co. Offaly)	6	211	-1.09901609547	0.273019413515	1	0.775205969578	No	4	-1.27485113322	0.203770651136	1	0.806936687333	No	
Edenderry (Co. Offaly)	12	211	0.285939959313	0.775205969578	1	0.775205969578	No	4	0.244688328193	0.806936687333	1	0.806936687333	No	
Edenderry (Co. Offaly)	24	210	0.797665284604	0.425970086924	1	0.775205969578	No	4	0.7202692372	0.472164006214	1	0.806936687333	No	
Henry St. Limerick	1	211	0.917485000076	0.359941046441	1	0.479921395255	No	4	0.817116492146	0.414787897338	1	0.553050529784	No	
Henry St. Limerick	6	211	-1.24178037355	0.215702715614	1	0.456770694829	No	4	-1.40751577817	0.160752949782	1	0.462213206051	No	
Henry St. Limerick	12	210	0.470496603328	0.638490888815	1	0.638490888815	No	4	0.44596391138	0.656084967302	1	0.656084967302	No	
Henry St. Limerick	24	211	1.20805352336	0.228385347415	1	0.456770694829	No	4	1.20099341271	0.231106603025	1	0.462213206051	No	
Portlaoise (Co. Laois)	1	212	1.22101525263	0.223442872238	1	0.363908377209	No	4	1.20506722449	0.229527144141	1	0.376582394821	No	
Portlaoise (Co. Laois)	6	212	-1.09920494905	0.272931282907	1	0.363908377209	No	4	-1.07761334803	0.282436796115	1	0.376582394821	No	
Portlaoise (Co. Laois)	12	212	1.66861164386	0.0966775319893	1	0.363908377209	No	4	1.85432754749	0.065087759069	1	0.260351036276	No	
Portlaoise (Co. Laois)	24	212	0.679846930458	0.497346576424	1	0.497346576424	No	4	0.703695267802	0.482398379991	1	0.482398379991	No	